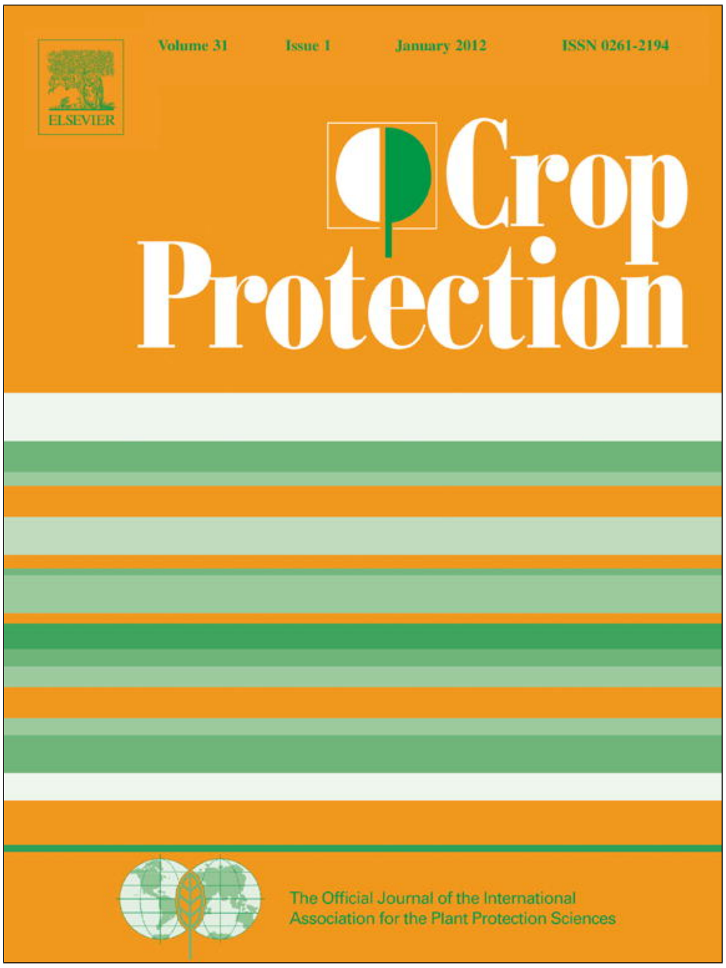
Applications of Nanomaterials in Agricultural Production and Crop Protection: A Review

Article *in* Crop Protection · May 2012

DOI: 10.1016/j.cropro.2012.01.007


Volume 31
Issue 1
January 2012
ISSN 0261-2194
ELSEVIER
Crop Protection
The Official Journal of the International Association for the Plant Protection Sciences

Crop Protection 35 (2012) 64–70

Contents lists available at SciVerse ScienceDirect

Crop Protection

journal homepage: www.elsevier.com/locate/cropro

Review

Applications of nanomaterials in agricultural production and crop protection: A review

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ARTICLE INFO

Article history:
Received 6 September 2011
Received in revised form
13 January 2012
Accepted 16 January 2012

Keywords:
Plant germination
Nanopesticides
Pesticide residue detection
Pathogen detection

ABSTRACT

Recent manufacturing advancements have led to the fabrication of nanomaterials of different sizes and shapes. These advancements are the base for further engineering to create unique properties targeted toward specific applications. Historically, various fields such as medicine, environmental science, and food processing have employed the successful and safe use of nanomaterials. However, use in agriculture, especially for plant protection and production, is an under-explored area in the research community. Preliminary studies show the potential of nanomaterials in improving seed germination and growth, plant protection, pathogen detection, and pesticide/herbicide residue detection. This review summarizes agricultural applications of nanomaterials and the role these can play in future agricultural production.

1. Introduction

Materials with a particle size less than 100 nm in at least one dimension are generally classified as nanomaterials. The development of nanotechnology in conjunction with biotechnology has significantly expanded the application domain of nanomaterials in various fields. A variety of carbon-based, metal and metal oxide-based dendrimers (nano-sized polymers) and biocomposites nanomaterials (EPA [Environment Protection Agency], 2007; Nair et al., 2010) are being developed. Types include single-walled and multi-walled carbon nanotubes (SWCNT/MWCNT), magnetized iron (Fe) nanoparticles, aluminum (Al), copper (Cu), gold (Au), silver (Ag), silica (Si), zinc (Zn) nanoparticles and zinc oxide (ZnO), titanium dioxide (TiO_2), and cerium oxide (Ce_2O_3), etc. General applications of these materials are found in water purification, wastewater treatment, environmental remediation, food processing and packaging, industrial and household purposes, medicine, and in smart sensor development (Jain, 2005; Wei et al., 2007; Chau et al., 2007; Byrappa et al., 2008; Zhang and Webster, 2009; Gao and Xu, 2009; Qureshi et al., 2009; Lee et al., 2010; Zambrano-Zaragoza et al., 2011; Bradley et al., 2011). The majority of applications in these areas have focused on the significance of the nanomaterials for improved efficiency and productivity. These materials are also used in agriculture production and crop protection (Bouwmeester et al., 2009; Nair et al., 2010; Sharon et al., 2010; Emamifar et al., 2010).

A precedent exists for conducting comprehensive literature reviews as a guide to the further development of nanomaterials applications. Reviews are available involving water disinfection (Li et al., 2008), the food industry (Sanguansri and Augustin, 2006), non-point source pollution control (Shan et al., 2009), treatment of environmental waste (Macaskie et al., 2010), and the design of trace concentration detection devices (Zhang and Fang, 2010). However in the field of agriculture, the use of nanomaterials is relatively new and needs further exploration. No previous literature reviews exist. As such, this article summarizes the developments and application of novel nanomaterials in agriculture. Topics include: plant germination and growth, plant protection and production, pathogen detection, and pesticide/herbicide residue detection.

2. Plant germination and growth

In recent years, various researchers have studied the effects of nanomaterials on plant germination and growth with the goal to promote its use for agricultural applications. Zheng et al. (2005) studied the effects of nano and non-nano TiO_2 on the growth of naturally-aged spinach seeds. It was reported that nano-TiO_2 treated seeds produced plants that had 73% more dry weight, three times higher photosynthetic rate, and 45% increase in chlorophyll-a formation compared to the control over germination period of 30

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doi:10.1016/j.cropro.2012.01.007

days. The growth rate of spinach seeds was inversely proportional to the material size indicating that smaller the nanomaterials the better the germination. The key reason for the increased growth rate could have been the photo-sterilization and photo-generation of “active oxygen like superoxide and hydroxide anions” by nano-TiO_2 that can increase the seed stress resistance and promote capsule penetration for intake of water and oxygen needed for fast germination. The authors concurred that the nano size of TiO_2 might have increased the absorption of inorganic nutrients, accelerated the breakdown of organic substances, and also caused quenching of oxygen free radicals formed during the photosynthetic process, hence increasing the photosynthetic rate.

The key to increased seed germination rate is the penetration of nanomaterials into the seed. Khodakovskaya et al. (2009) reported that MWCNTs can penetrate tomato seeds and increase the germination rate by increasing the seed water uptake. The MWCNTs increased the seed germination, up to 90% (compared to 71% in control) in 20 days, and the plant biomass. However, authors insisted the importance of additional studies for evaluating the resistance of CNT germinated tomato plants to pests and also evaluating the toxic effects of CNT on other field plants prior to their direct field applications. Study on the influence of metal nanoparticles (Si, Palladium-Pd, Au, Cu) on germination of lettuce seeds (Shah and Belozerova, 2009) indicated that nanoparticles (Pd, Au at low concentrations; Si, Cu at higher concentrations, and combination of Au and Cu) had a positive influence on seed germination, measured in terms of shoot to root ratio and growth of the seedling. Authors also attempted to determine if the nanoparticles affected the soil microorganisms, but no definite effect could be found.

The effect of nanoparticles on plants can be positive or negative (Monica and Cremonini, 2009). One of the concerns for nanomaterials applications in seed germination is their phytotoxicity. The level of phytotoxicity may depend on the type of nanomaterial and its potential application. For example, the applicability of fluorescein isothiocyanate (FTIC)-labeled silica nanoparticles and photostable Cadmium-Selenide (CdSe) quantum dots were tested for their ability to be used as biolabels and for promoting seed germination. It was found that FTIC-labeled silica nanoparticles induced seed germination in rice, while quantum dots arrested the germination (Nair et al., 2011).

Lin and Xing (2007) evaluated phytotoxicity of nanomaterials (MWCNTs, Aluminum oxide-Al_2O_3, ZnO, Al and Zn) and its impact on germination rates in radish, rape canola, ryegrass, lettuce, corn, and cucumber. They conferred the hypothesis that the higher concentrations (2000 mg/L) of nano-sized Zn (35 nm) and ZnO ($\sim$20 nm) inhibited the germination in ryegrass and corn, respectively. Root length of studied species was also inhibited with use of 200 mg/L nano-Zn and ZnO. Phytotoxicity of nano-Al and Al_2O_3 significantly affected root elongation of ryegrass and corn, respectively; whereas, nano-Al facilitated the radish and rape root growth.

Ma et al. (2010) studied the effects of four oxide nanoparticles (CeO_2, Lanthanum (III) oxide-La_2O_3, Gadolinium (III) oxide-Gd_2O_3, Ytterbium oxide-Yb_2O_3) on the radish, rape, tomato, lettuce, wheat, cabbage, and cucumber plant species. Similar to Lin and Xing (2007), they found that the root growth depended on nanoparticles and its concentration. Ma et al. (2010) reported that the nano-CeO_2 did not affect root elongation in plant species except for lettuce at 2000 mg/L concentration. However, the other three types of nanoparticles (La_2O_3, Gd_2O_3, Yb_2O_3) greatly affected root growth at same concentration. Moreover, the inhibitory effect of these nanoparticles was observed during different stages of root growth.

Thus, the phytotoxic behavior of the nanomaterials needs to be thoroughly understood before utilizing nanomaterials under field conditions. A possible solution to avoid the phytotoxicity to other plant species would be to grow the plant seedlings in a greenhouse and later transferring them to field. This would be more suitable for ornamental and specialty crops.

Applicability and phytotoxicity of silver nanomaterials in agriculture is well debated by the EPA (Bergeson, 2010a,b). As reported by Bergeson (2010a), there are more than 100 pesticides that contain Ag due to its anti-microbial properties. However, toxicity of nanosilver to ecosystem and human is a major concern. Lu et al. (2010) have reported that the citrate-coated colloidal Ag nanoparticles were not genotoxic (genetic), cytotoxic (cell), and phototoxic (toxicity through photo-degradation) to humans, however; citrate-coated Ag nanoparticles in powder form were toxic. The authors argued that this could be because of the “chemical change of spherical silver nanoparticle in the powder to form silver oxides or ions.” Interestingly, the photoxicity of the powdered Ag nanoparticles was repressed by coating them with biocompatible polyvinylpyrrole (Lu et al., 2010). Exploring such biocompatible coatings to reverse the toxicity of nanomaterials would increase the chances of applying nanomaterials in plant germination and growth. Research is also needed to investigate the adverse effect of such coatings on the desired seed/plant properties and the effectiveness of nanomaterials.

Oancea et al. (2009) hypothesized that controlled release of active plant growth stimulators and other chemicals encapsulated in nanocomposites made of layered double hydroxides (anionic clays) could be another feasible option for organic agriculture. However, leading food organic certifiers (e.g. UK soil association, Biological farmers of Australia) abstain from considering nanomaterial-based agri-foods as of organic standards (Scrinis and Lyons, 2010). Recently, German-based organizations such as Naturland and the International Federation of Organic Agriculture Movements (IFOAM) prohibited labeling food products grown with artificial nanomaterials as organic (Naturland, 2011; IFOAM, 2011).

Nonetheless, future research on nanomaterials for plant germination and growth should address some of the following challenges (as reported by Nair et al., 2010): 1) unpredictability in reaction of nanomaterials to different plants, 2) phytotoxicity due to higher concentrations, and 3) reduced intake and photosynthesis of plant due to larger nanomaterials.

3. Plant protection and production

Nanopesticides “involve either very small particles of pesticidal active ingredients or other small engineered structures with useful pesticidal properties” (Bergeson, 2010b). Nanopesticides can increase the dispersion and wettability of agricultural formulations (i.e., reduction in organic solvent runoff), and unwanted pesticide movement (Bergeson, 2010a). Nanomaterials and biocomposites exhibit useful properties such as stiffness, permeability, crystallinity, thermal stability, solubility, and biodegradability (Bouwmeester et al., 2009; Bordes et al., 2009) needed for formulating nanopesticides. Nanopesticides also offer large specific surface area and hence increased affinity to the target (Jianhui et al., 2005). Nanoemulsions, nanoencapsulates, nanocontainers, and nanocages are some of the nanopesticide delivery techniques that have been discussed recently (Lyons and Scrinis, 2009; Bouwmeester et al., 2009; Bergeson, 2010b) for plant protection. Table 1 reports some of the recent applications of nanomaterials in agricultural plant protection and production.

Basically, the nano-formulations should degrade faster in the soil and slowly in plants with residue levels below the regulatory criteria in foodstuffs. Jianhui et al. (2005) reported the development of such sodium dodecyl sulfate (SDS) modified photocatalytic TiO_2/Ag nanomaterial conjugated with dimethomorph

Table 1
Nanomaterials in agricultural plant protection and production.

Purpose	Material	Findings	Reference
Smart agrochemical delivery system via plant roots of sunflower, tomato, pea and wheat	Magnetic carbon coated nanoparticles	Nanoparticles moved through plant xylem and phloem within 24 h	Cifuentes et al., 2010
Controlled release herbicide delivery system for atrazine	Polyhydroxybutyrate-co-hydroxyvalerate microspheres with atrazine (~13 nm)	Good affinity of herbicide with polymer, decreased genotoxicity and increased biodegradability	Grillo et al., 2010
Nanocomposite based controlled release of herbicide, 2,4-dichlorophenoxyacetate (2,4-D)	Inorganic Zn–Al layered double hydroxide (ZAL) as release agent	Initial burst of 2,4-D followed by sustained release that depended on type of anions and their concentrations in release medium	Hussein et al., 2005
Controlled delivery system for water-soluble pesticide (validamycin)	Porous hollow silica nanoparticles (PHSNs)	Pesticide was loaded into PHSNs (36 wt % loading capacity) and the release was in two stages: initial burst followed by sustained release	Liu et al., 2006
Reduce the bean rust disease severity	CNT conjugated with INF24 oligonucleotides	Treatment reduced the rust severity	Corrêa et al., 2010
Control of lentil pathogen and wilting	Silver nanoparticles-AgNPs (0.5–1000 ppm)	Faster plant growth compared to control; AgNPs did not reduce the plant wilting	Ashrafi et al., 2010
Physical and biological changes of *Brassica oleracea* in presence of nanomaterials	TiO_2 (5–8 nm) 0.05–2 mL of TiO_2 in 500 mL of Hoagland solution	Higher concentrations had negative impact on shoot length whereas positive impact on root length	Singh et al., 2010
Effect of carbon nanostructures on tomato germination	MWCNTs	Seed germination was not related to MWNCTs (observed up to 7 days)	Lima et al., 2010
Treatment of fungal pathogens *in vitro* and in chickpea and wheat plants	Amphotericin B nanodisks (AMB-NDs) 0.1–2 μg/mL (*in vitro*), 0.1–10 μg/L (plants)	AMB-NDs inhibited fungi at 0.1 μg/mL (*in vitro*); effective chickpea fusarium wilt control (preventive dosage of 0.1 μg/L), wheat leaf rust control by foliar treatment	Perez-de-Luque et al., 2012

(DMM), commonly used pesticide in agricultural production, as a nanopesticide. Modified formulation, 96 nm average granularity, increased dispersivity and decomposition of the pesticide in soil while increasing its effectiveness in vegetable seedling (of cabbage and cucumber) studies. The modification of the nanomaterials using SDS greatly increased the absorption of the DMM. Recently, Guan et al. (2010) formulated encapsulated nano-imidacloprid with above properties that could be used for pests control during vegetable production. The SDS modified Ag/TiO_2 imidacloprid nanoformulation was developed using a microencapsulation technique that used chitosan and alginate. It was tested on soybean plants that were transplanted to soil with 3.1% dry matter content and pH of 6.2. The formulation residues in soil and the plants degraded faster during the first eight days, and were minimal to undetectable after 20 days.

The SDS in the above applications was used to increase the photo-degradation of the nanoparticles in soil. Alternatively, Mohamed and Khairou (2011) developed highly photo-degradable Ag/TiO_2 particles (5–7 nm), synthesized using polyoxyethylene laurel ether (POL). They tested Ag/TiO_2 nanoparticles synthesized using both POL and SDS toward the degradation of 2,4-D herbicide under visible and UV radiation. Results confirmed that the POL synthesized nanoparticles photo-degraded faster during the same exposure period.

All the above results cannot be considered as safeguarded due to lack of any national/international nano-regularity institutional (Bowman and Hodge, 2007) framework/standards related to the nanomaterials agriculture use paradigm. Toxicity or biosafety of pesticides is another major concern in agricultural production. With nanopesticide applications, the uncertainty on the long-term effects of pesticides on the human health and environment increases. Xu et al. (2010) stated that with better kinetic stability, smaller size, low viscosity and optical transparency, nanoemulsions can potentially be better pesticide delivery medium. The micro or nanoemulsion as a carrier for pesticide delivery can improve the solubility and bioavailability of nanopesticides. However, there is a need to assess the possible uptake of nanopesticides that can occur when agricultural workers inhale it during applications.

Shi et al. (2010) studied the toxicity of chlorfenapyr (nanopesticide) on mice. It was reported that the chlorfenapyr nanoformulation from 4.84 mg/kg to 19.36 mg/kg was less toxic to mice than the common formulation. Thus, such nanoformulation pesticides may decrease adverse environmental and human effect as compared to normal pesticide application. In order to ensure health and environmental safety, further such studies on the biosafety of the nanopesticides are needed.

Formulation stability is also an important aspect at the nano level. Liu et al. (2008) successfully formulated a stable nanopesticide (bifenthrin) using polymer stabilizers such as Poly(acrylic acid)-b-poly(butylacrylate) (PAA-b-PBA), Polyvinylpyrrolidone (PVP), and Polyvinyl alcohol (PVOH). A flash nano-precipitation technique was used to prepare 60–200 nm bifenthrin particles. While using such techniques commercially, stability of the polymers over an extended period of time needs to be considered. Although not for agriculture, Anjali et al. (2010) reported formulation of artificial polymer-free nanopermethrin as an effective larvicide that was stabilized by plant extracted natural surfactants. More of such studies that investigate the use of natural stabilizers for nanopesticide formulations in agricultural plant protection are needed. Another area of advanced research could be the development of nanomaterials that can be used as a coating or protective layer to enable slow release of traditional pesticides and fertilizers. For example, Corradini et al. (2010) explored the possibility of utilizing chitosan nanoparticles, a highly degradable antibacterial material for slow release of NPK fertilizer. Liu et al. (2006)

developed kaolin clay-based nanolayers to be used as cementing and coating material for control release of fertilizers. More of such studies have been summarized in Ghormade et al. (2010). Primarily, nano-clay materials offer interactive surfaces with high aspect ratio for encapsulating "agrochemicals such as fertilizers, plant growth promoters, and pesticides" (Ghormade et al., 2010).

4. Pesticide residue detection

There are about 1045 chemicals reported by Food and Drug Administration (FDA) as pesticide residues (FDA, 2005). Nanomaterials based nanosensors can be used to detect such pesticide residues as an alternate to traditional gas or liquid chromatography (GC/LC) -mass spectroscopy (-MS) techniques (Stan and Linkerhägner, 1996; Sicbaldi et al., 1997; Balinova et al., 2007). Although accurate and reliable, above traditional techniques involve time consuming steps such as: field sample collection, solid-phase extraction in laboratory, analyzing the sample, and comparing the obtained spectral peaks with references to determine the pesticide residues. Currently, U.S. Department of Agriculture (USDA) follows single- and multi-residue methods based GC/LC-MS to evaluate "organophosphates, organochlorines, carbamates, triazines, triazoles, pyrethroids, neonicotinyls, strobilurins" residues in 85 agricultural commodities (USDA, 2010). USDA-Pesticide Data Program (PDA) analyzes more than 10,000 samples a year to determine the pesticide residues (USDA-PDA, 2008) in agricultural related commodities. Considering above vastness, an alternative to analytical methods could not only increase the sampling rate but also reduce the cost associated with analytical methods.

Nanosensors for pesticide residue detection offer, "high sensitivity, low detection limits, super selectivity, fast responses, and small sizes" (Liu et al., 2008). Table 2 reports some of the nanosensors aimed to detect the pesticide residues such as methyl parathion (Kang et al., 2010; Parham and Rahbar, 2010), parathion (Li et al., 2006; Wang and Li, 2008), fenitrothion (Kumaravel and Chandrasekaran, 2011), pirimicarb (Sun and Fung, 2006), and dichlorvos and paraoxon (Vamvakaki and Chaniotakis, 2007).

Additionally, Dyk and Pletschke (2011) have reviewed enzyme-based biosensors for organochlorines, organophosphates, and carbamates residue detection. Some of these biosensors used C, Au, hybrid Titanium (Ti), Au-Platinum (Pt), and nanostructured lead dioxide (PbO_2)/TiO_2/Ti to immobilize the enzymes on sensor substrate and to increase the sensor sensitivity. Application domain of nanomaterials based sensing for pesticide residue detection is vast, nevertheless some issues such as: 1) availability of the nanomaterials sensitive to common pesticide residues, 2) ease of sensor fabrication techniques and instrumentation, 3) desired reliability and repeatability in trace level detection, 4) cost, and 5) issues related to nanomaterial exposure to the surrounding environment needs to be considered. Also, nanomaterials based sensors applicability for pesticide residue detection might be restricted due to the large number of pesticides used in agriculture production (Liu et al., 2008; Dyk and Pletschke, 2011). It would be "impossible to test randomly for all pesticides in samples" (Dyk and Pletschke, 2011). Moreover, as reported by Liu et al. (2008), the advancement in the development of selective and stable nanomaterial sensing and techniques to integrate biomolecules (enzymes, antibodies, etc.) with nanomaterials is needed. As a starting point, the regulatory institutions can use nanosensors approach to detect major residuals that are extremely harmful to human health. Smart nanomaterials can also be used as an alternative to sensors for pesticide detection. The smart nanomaterials and nanopesticides (Bergeson, 2010b) that act as a source of pesticide as well as indicative sensor could eliminate the need of sensors for detecting pesticide residues in soil. The nanomaterial that would have slow targeted release of the material and also indicate deficiency (e.g. color change) of the nutrients in soil would work as an advanced alert system for farmers to decide upon the dosage rate and frequency.

Nanomaterials, in addition to its use for pesticide and herbicide detection, have also been applied for pesticide degradation. Yu et al. (2007) studied the potential of using nano-TiO_2 film in photocatalytic degradation of organochlorine pesticides. They stated that the attraction of peroxide or hydroxyl radicle and electron transfer enables the photolytic degradation of pesticides on the surface of TiO_2. Zeng et al. (2010) established that organophosphorus and carbamate pesticides in tomato leaves and soil can also be photocatalytically degraded at a rate of 15–30% using a Rhenium (Re^{3+})-doped nano TiO_2. Same nanomaterial was capable for

Table 2
Nanomaterials for pesticide residue detection.

Pesticide/herbicide detection	Sensing material	Detection limit	Sensor type	Reference
2,4,5-trichlorophenoxy acetic acid	Poly-o-toluidine Zirconium (IV) phosphate nanocomposite	1 μM	Electrochemical	Khan and Akhtar, 2011
Fenitrothion in water	Nano TiO_2/nafion composite	0.13 μM	Electrochemical	Kumaravel and Chandrasekaran, 2011
Melamine in milk	18-crown-6 ether functionalized Au nanoparticles	6 ppb	Optical	Kuang et al., 2011
Organochlorine and organophosphorus pesticides in cabbage	Amino-functionalized nanocomposite with tetraethylenepentamine	0.29 μg/kg	Chromatography	Zhao et al., 2011
Methyl parathion in water	Nano-ZrO_2/graphite/Paraffin	2 ng/mL	Electrochemical	Parham and Rahbar, 2010
Methyl parathion in vegetables (cabbage, spinach, lettuce)	Nano-Au/Nafion composite	10^{-7} M	Electrochemical	Kang et al., 2010
Methyl parathion, chlorpyrifos	Nano size polyaniline matrix with SWCNT, single stranded DNA and enzyme	1 pM	Electrochemical biosensor	Viswanathan et al., 2009
Fenamithion and acetamiprid in water	Cd-Tellurium quantum dots with *p*-sulfonatocalix[4]arene	0.12 and 0.34 nM	Luminescence	Qu et al., 2009
Parathion in water	Nano-ZrO_2/Au composite	3 ng/mL	Electrochemical	Wang and Li, 2008
Dichlorvos and paraoxon in drinking water	Acetyl cholinesterase and Pyranine immobilized on nano-sized liposomes	10^{-10} M	Optical Nano-biosensor	Vamvakaki and Chaniotakis, 2007
Parathion in vegetables	Nano-TiO_2 on glassy carbon electrode	10^{-8} M	Electrochemical	Li et al., 2006
Pirimicarb in vegetables	Molecular imprinted nano-polymers (methacrylic acid with carboxyl functional groups)	8×10^{-6} M	Piezoelectric	Sun and Fung, 2006

degrading carbofuran at a rate of 55%, which is 30% higher than natural degradation.

5. Plant pathogen detection

"Smart field systems detect, locate, and report on pathogens, then apply pesticides and fertilizers as needed prior to the onset of symptoms" (Bergeson, 2010a). Intuitively, nanoparticles can be used as biomarkers or as a rapid diagnostic tool for detection of bacterial, viral and fungal plant pathogens (Boonham et al., 2008; Yao et al., 2009; Chartuprayoon et al., 2010) in agriculture, though this research is in preliminary stage. Nanoparticles based sensors might offer improved detection limits in detecting viral pathogens in plant (Baac et al., 2006). Nanoparticles can either be directly modified to be used for pathogen detection, or used as a diagnostic tool to detect compounds indicative to a diseased condition. Nano-chips are types of microarrays that contain fluorescent oligo capture probes through which the hybridization can be detected (López et al., 2009). Above nano-chips are known for their sensitivity and specificity in detecting single nucleotide changes of bacteria and viruses (López et al., 2009). Yao et al. (2009) utilized a fluorescence silica nanoparticles in combination with antibody to detect *Xanthomonas axonopodis* pv. *vesicatoria* that causes bacterial spot disease in Solanaceae plants, indicating a potential for nanoparticle application in disease detection. Singh et al. (2010) used nano-gold based immunosensors that could detect Karnal bunt (*Tilletia indica*) disease in wheat using surface plasmon resonance (SPR). Particularly, research attempted to detect the disease using SPR sensor in wheat plots for seed certification and to establish plant quarantines. Research on pathogen detecting nanosensors for their in-field application would be highly valuable for rapid diagnosis and disease management.

Plants respond to different stress conditions through physiological changes. One such response is the induction of systemic defense, which is thought to be regulated by plant hormones: jasmonic acid, methyl jasmonate and salicylic acid. Wang et al. (2010) harnessed this indirect stimulus to develop a sensitive electrochemical sensor, using modified gold electrode with copper nanoparticles, to monitor the levels of salicylic acid in oil seeds to detect the fungi (*Sclerotinia sclerotiorum*). They successfully and accurately measured salicylic acid using this sensor. Research on similar sensors and sensing techniques needs to be expanded for detecting pathogens, their byproducts, or monitor physiological changes in plants.

6. Issues and future research needs

Toxicity of the ecosystem (Illuminato, 2009), potential residue carry-over in foodstuff (Chaudhari and Castle, 2011), and nanomaterials phytotoxicity are some of the major concerns for application of nanomaterials in agriculture. Bouwmeester et al. (2009) have reviewed in-detail the health concerns of nanomaterials related to plant production. Kahru and Dubourguier (2010) have also discussed the health hazards of various nanomaterials. Additionally, there is a need to evaluate the "toxicokinetics and toxicodynamics" of nanomaterials (Bouwmeester et al., 2009; Bergeson, 2010a) used for agricultural production. Nair et al. (2010) highlighted that the same nanomaterials have different effect on various agricultural plants and will affect considerably the alternative cropping systems if not degraded quickly. Studies on such aspects needs to be conducted and their implications need to be understood thoroughly before using the nanomaterials in agricultural production system. Site-specific application technologies of nanomaterials would somewhat reduce such risks. Major issues related to the application of nanomaterials in agricultural production and crop protection are (summarized from FAO/WHO meeting report, 2010): a) accurate characterization of nanomaterials in biological matrices for in-depth understanding of their toxicity in biological systems, b) nanomaterials interaction with biology, c) dose–response considerations, d) exposure assessment and characterization studies, e) product life cycle considerations, f) nanomaterials background levels in food and feed matrices, and g) nanomaterial amount and form in foodstuff due to their use in agricultural production and crop protection. Additionally, IFPRI [International Food Policy Research Institute] (2011) policy brief recommends to "conduct risk analysis so decision makers understand the cost effectiveness of using certain nanotechnology applications to improve food and water safety compared to other technologies".

Like other technologies, low-cost nanomaterials and field application technologies are needed for their applications in agriculture. Also, the current application methods need to be reviewed for increased efficacy of nanomaterials/nanopesticides on intended targets. Similar to pesticide drift assessment models, there is a need for models to predict the nanomaterial exposure limits and scenarios with details of transport, transformation and toxicity assessment of each particular nanomaterial. As reported by Navarro et al. (2008), the detailed studies on nanomaterials aspects such as: the nanoparticles dosage in the different environments, their physical and chemical characterization, the mechanisms allowing them to pass through cellular membranes and cell walls, the specific properties that are related to toxic effects of nanoparticles, and the mechanism underlying nanoparticles trophic transfers needs to be conducted. Hazard of nanomaterials to human health has been highlighted by many researchers including Reijnders (2006) and Suh et al. (2009). Inhalation, ingestion, and dermal exposure are the key uptake routes of nanomaterials in human body (Xu et al., 2010). Xu et al. (2010) also highlighted the need to investigate factors such as the size, chemical composition, surface structure, solubility and aggregation of the nanomaterials for their associated exposure risk effects to humans. Efficient application technologies of the nanomaterials in agricultural production would reduce this exposure risk. In summary, the development of nanomaterials: with good dispersion and wettability, biodegradable in soil and environment, less toxic and more photo-generative, with well understood toxicokinetics and toxicodynamics, smart and stable, and ease of fabrication and application in agriculture, would be ideal for their effective use in agricultural production.

7. Summary

This review has focused on specific applications of nanomaterials in agriculture such as plant protection, pathogen detection, and pesticide residue detection, amongst others. Applications of nanomaterials can help faster plant germination/production, effective plant protection with reduced environmental impact as opposed to traditional approaches. Also, smart nanosensors could be a viable option to detect pesticide residues in the field. This review has also discussed some of the issues and concerns in regard to the specific nanomaterial applications, and a few possible solutions to these issues. For example, nanomaterials can improve plant germination in certain plants; while adversely affecting other plants. In such cases, the nanomaterials can be applied under controlled conditions (such as in greenhouse-grown plants) to promote germination in the plants of interest. Although this review demonstrates the potential of nanomaterials for different agricultural applications, further investigation and research is needed to expand the application possibilities and methodologies in agriculture.

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